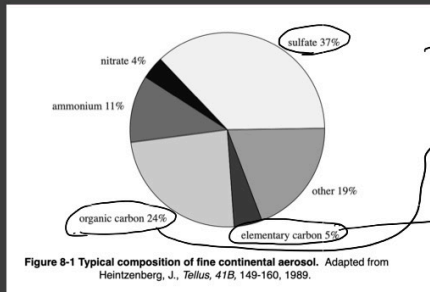


AEROSOLS → origin from condensation of gases & action of wind on earth's atmosphere.

FINE aerosol → $< 1 \mu\text{m}$ in radius → originate almost exclusively from condensation of precursor gases.



for lower troposphere. One key gas is H_2SO_4 → formed by SO_2 emitted by volcanoes, fuel combustion etc.
 H_2SO_4 → Low vapour pressure → condenses easily
 condensation of large hydrocarbons.
 major mentioning of soot → formed by condensation of gas during combustion.
 form of aqueous sulphate particles
 * act as CCM → cloud condensation nuclei

Action of wind → sea salts soil dust vegetation debris → into atmosphere

size = $1-10 \mu\text{m}$ in radius.
 Now
 $< 1 \mu\text{m}$ difficult to generate mechanically because of high Area/Volume & high surface tension
 $> 10 \mu\text{m}$ not lifted by wind that easily & very high sedimentation velocities.
 * So very low atmospheric life

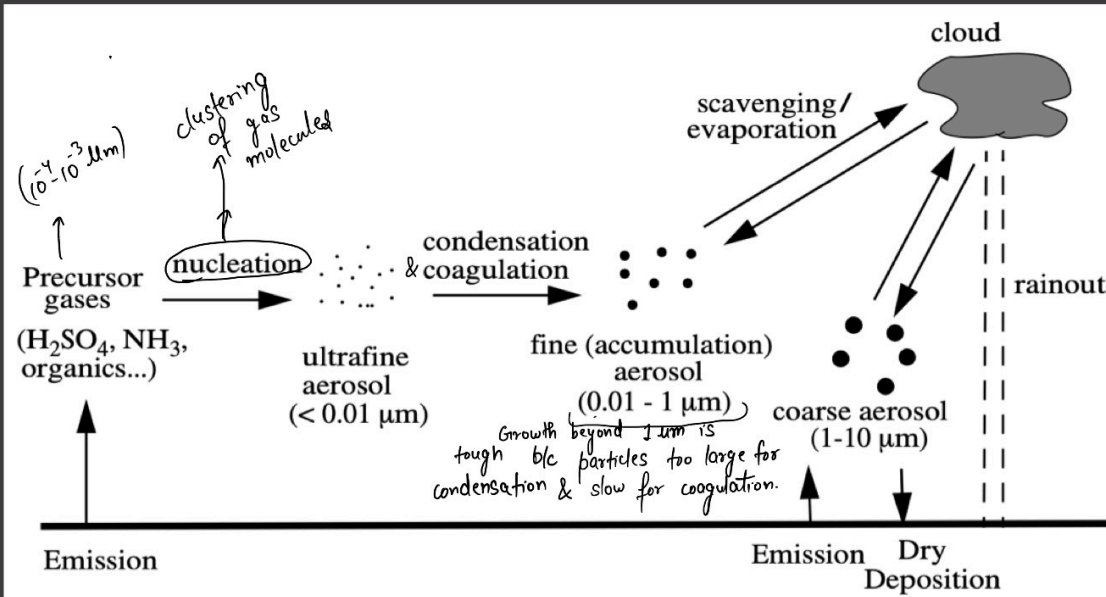


Figure 8-2 Production, growth, and removal of atmospheric aerosols

$(0.01 - 1 \mu\text{m})$ → accumulation mode
 ↓
 too small to sediment at significant rate
 * so removed by scavenging by water droplets and rainout.
 but coarse aerosol can be removed by sedimentation along with this.
 → bulk aerosols in lower troposphere b/c of deposition

Stratosphere contains $\text{H}_2\text{SO}_4 - \text{H}_2\text{O}$ layer at 15-25 km which plays an important role for ozone chemistry
 Arises mainly from Carbonyl Sulphide (COS)

* # Lifetime (aerosols) in Stratosphere > troposphere due to lack of precipitation.

Scattering $\rightarrow$ dirⁿ of propagation altered without any absorption taking place.
 Scattering efficiency of smaller ones $\ll$ larger ones of accumulation mode } for solar radiations
 Soot $\rightarrow$ absorbs radiations

• Scattering max when $\lambda_{\text{light}} \approx r_{\text{particle}}$ *

reason: diffraction becomes significant

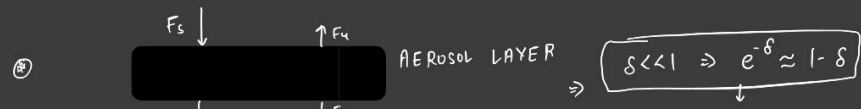
Scattering increases Earth's albedo

- Aerosols cause $\rightarrow$ visibility redⁿ $\rightarrow$ In absence of aerosols, our visible range could be 300 km
- Visibility redⁿ $\rightarrow$ highest at high relative humidities
 * particles swell up by uptake of water

Climate effect by aerosol layer of optical depth (δ)

$$\Rightarrow \Delta F = \frac{-F_s \Delta A}{4} \rightarrow \text{Increase in Earth's albedo}$$

* anthropogenic aerosols contribute 25% of aerosols
 $\delta = 0.1$



$$F_t = F_s e^{-\delta}$$

$$\text{Total scattered flux} = f_u + f_d = F_s - F_t = F_s - F_s(1 - \delta) = F_s \delta$$

Albedo (A^*) for aerosol layer

$$A^* = \frac{F_u}{F_s}$$

Now most scattering $\rightarrow$ instead of $\leftarrow$
 $\Rightarrow$ backward = β = 0.9 fraction of total

$$\text{So } F_u = \beta (F_d + F_u)$$

$$\delta F_s \Rightarrow F_u = \delta \beta F_s$$

$$\Rightarrow A^* = \frac{\delta \beta F_s}{F_s} = \delta \beta \approx \frac{0.1}{10} \times \frac{1}{10} \times \frac{1}{4} \times \frac{10}{10} \approx 5 \times 10^{-3} \text{ for anthropogenic while } 20 \times 10^{-3} \text{ for all aerosol}$$

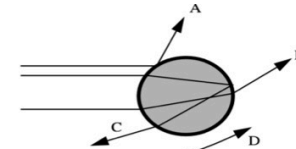


Figure 8-3 Scattering of a radiation beam: processes of reflection (A), refraction (B), refraction and internal reflection (C), and diffraction (D).

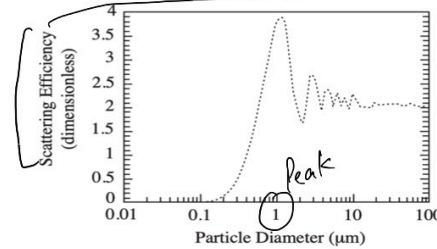


Figure 8-4 Scattering efficiency of green light ($\lambda = 0.5 \mu\text{m}$) by a liquid water sphere as a function of the diameter of the sphere. Scattering efficiencies can be larger than unity because of diffraction. Adapted from Jacobson, M.Z., *Fundamentals of Atmospheric Modeling*, Cambridge University Press, Cambridge, 1998.

$Q_{\text{scat}} > 1$ means particle as per its surface area, aurat se jyada scattering kar rha hai wave nature of light involve ho jata hai

SOLAR RADIATION

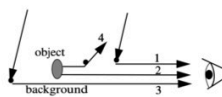


Figure 8-5 Reduction of visibility by aerosols. The visibility of an object is determined by its contrast with the background (2 vs 3). This contrast is reduced by aerosol scattering of solar radiation into the line of sight (1) and by scattering of radiation from the object out of the line of sight (4).

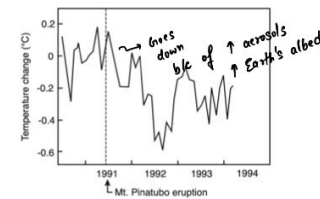
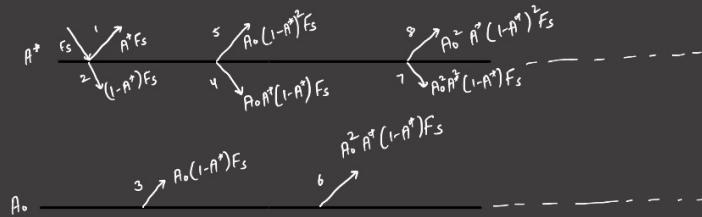


Figure 8-6 Observed change of the Earth's global mean surface temperature following the Mt. Pinatubo eruption (September 1991). Adapted from *Climate Change 1994*, Cambridge University Press, New York, 1995.

Though $\Delta A < A^*$
 ↓
 Horizontal overlap with reflective surfaces already



Let A_T = Total Albedo from superimposed layers A^* & A_0

$$A_T = \left[A^* + A_0(1-A^*)^2 + A_0^2A^*(1-A^*)^2 + \sum_{n=2}^{\infty} A_0^{n+1}A^*(1-A^*)^2 \right] F_s$$

$\Rightarrow A^* \ll 1 \rightarrow$ neglect terms with $n \geq 2$

$$\text{So } A_T = A^* + A_0(1-A^*)^2 + A_0^2A^*(1-A^*)^2$$

$$= A^* + A_0(1+A^*^2-2A^*) + A_0^2A^*(1+A^*^2-2A^*)$$

$$= A^* + A_0 + A_0A^*^2 - 2A_0A^* + A_0^2A^* + A_0^2A^*^3 - 2A_0^2A^*^2$$

$$\Rightarrow A_T = A_0 + A^*(1-2A_0+A_0^2) = A_0 + A^*(1-A_0)^2$$

$$\text{Now } \Delta A = A_T - A_0$$

$$= A^*(1-A_0)^2$$

$$\text{So } \Delta F = \frac{-F_s A^*(1-A_0)^2}{4} \approx 0.9 \text{ W m}^{-2}$$

⊗ There is an indirect effect by aerosols (CCN)
 ↓
 increases area for cloud forⁿ,
 so greater cloud albedo

⊗ Soil dust aerosol also have a high optical depth,
 b/c of the large size, they scatter the solar radiation
 but also absorb the terrestrial radⁿ.